Algebra 1
weeks 1-4

Def 1.1 A set M together with an operation $M \times M \to M$, $(a,b) \mapsto ab$ is called a monoid if the following properties are satisfied:
(i) Associativity: $(ab)c = a(bc)$, $\forall a,b,c \in M$
(ii) Neutral element: there exists an $e \in M$ with $em = m = me$ for all $m \in M$

Rem 1.2
1) e is uniquely determined. If e' is another neutral element, then $e = e \cdot e' = e'$.
2) By (i), parentheses can be omitted and for $m_1, \dots, m_n \in M$ we can speak of the product $m_1 \cdots m_n \in M$. For $a \in M$ and $n \in \mathbb{N}$, we set $a^n = a \cdots a$ (n times). By convention, $a^0 = e$.

Def 1.3 Let $(M, \cdot)$ be a monoid and $a \in M$. An element $b \in M$ is called inverse to a if $ab = e = ba$ holds.

Rem 1.4 b, if it exists, is uniquely determined by a: If b' is also an inverse, then $b = eb = b'ab = b'e = b'$.

Def 1.5 A monoid $(G, \cdot)$ is called a group if every element has an inverse.

Rem 1.6 This can be weakened to: associativity, existence of a left-neutral element, existence of left inverses, and then showing that the left-neutral element is also right-neutral and a left inverse is also a right inverse.

Def 1.8 Let G be a monoid. A subset $H \subset G$ is called a submonoid if
(i) $e \in H$,
(ii) $a, b \in H \Rightarrow ab \in H$.
If G is a group, H is called a subgroup if additionally
(iii) $a \in H \Rightarrow a^{-1} \in H$.

Def 1.7 A monoid (a group) is called commutative if $ab = ba$ holds for arbitrary a, b.

Def 1.15 Let G be a group and $H \subset G$ a subgroup. A left coset of H in G is a subset of the form $aH := \{ah \mid h \in H\}$.

Def 1.32 If $H_1, H_2 \subset G$ are subgroups, H_1H_2 denotes the set $H_1H_2 = \{h_1h_2 \mid h_1 \in H_1, h_2 \in H_2\}$.

Def 1.38 The intersection of all subgroups $H \subset G$ with $X \subset H$ is called the subgroup generated by X, denoted $\langle X \rangle$. We say G is generated by X if $\langle X \rangle = G$.

Def 1.9 Let G, G' be monoids with neutral elements e, e'. A map $\varphi : G \to G'$ is called a monoid homomorphism if
(i) $\varphi(e) = e'$.
(ii) $\varphi(ab) = \varphi(a)\varphi(b) \quad \forall a, b \in G$.

Def 2.1 A ring (with 1) is a set R with two binary operations $+$ and $\cdot$ and elements $0, 1 \in R$ such that
(i) $(R, +, 0)$ is a commutative group,
(ii) $(R, \cdot, 1)$ is a monoid,
(iii) $(a+b)c = ac + bc$ and $c(a+b) = ca + cb$ for all $a, b, c \in R$.
R is called commutative if multiplication is commutative. $S \subset R$ is a subring if $(S, +)$ is a subgroup and $(S, \cdot)$ is a submonoid. $f : R \to R'$ is a ring homomorphism if $f(a+b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$, and $f(1) = 1$. (Hereafter all rings are assumed commutative.)

Ex 1.7a $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ with $+$ are commutative groups. $(\mathbb{Z}, \cdot)$ and $(\mathbb{N}_0, +)$ are commutative monoids. $\mathfrak{S}_n = \mathrm{Aut}(\{1, \dots, n\})$ is a group, non-commutative for $n \geq 3$. If k is a field, the matrix groups $\mathrm{Gl}_n(k)$ and $\mathrm{Sl}_n(k)$ are groups, non-commutative for $n > 1$.

Rem 1.13 Let G be a commutative group and $n \in \mathbb{N}$. Then $G \to G, g \mapsto g^n$ is a group homomorphism.

Lem 1.17 Let aH, bH be two left cosets in G. Then the following are equivalent:
(i) $aH = bH$.
(ii) $aH \cap bH \neq \emptyset$.
(iii) $a \in bH$.
(iv) $b^{-1}a \in H$.

Def 1.22 A subgroup $H \subset G$ is called a normal subgroup if $aH = Ha$ for all $a \in G$ ($H \triangleleft G$). In this case, $aH = Ha$ is called the residue class of a modulo H.

Def 1.20 The index of H in G is the number of left (right) cosets of H in G, denoted $(G : H)$. The order of G is the number of elements of G, denoted $\mathrm{ord}(G)$.

Rem 1.40 If $X = \{x_1, \dots, x_r\}$ is finite, one also writes $\langle x_1, \dots, x_r \rangle$ instead of $\langle \{x_1, \dots, x_r\} \rangle$.

Rem 1.14 Let G be a group and $a \in G$. Then $\varphi_a : G \to G, g \mapsto aga^{-1}$, is a group homomorphism (inner automorphism). The map $G \to \mathrm{Aut}(G), a \mapsto \varphi_a$ is a group homomorphism, which is trivial if and only if G is commutative.

Rem 1.10 If G and G' are groups, (i) follows from (ii). Moreover, φ is injective if and only if its kernel $\ker(\varphi) = \varphi^{-1}(e')$ is trivial, i.e., equal to $\{e\}$ (since $\varphi(g) = \varphi(h) \Leftrightarrow gh^{-1} \in \ker(\varphi)$).

Lem 1.39 $\langle X \rangle$ is the subset of all elements of G of the form $x_1^{\epsilon_1} \cdots x_n^{\epsilon_n}$ with $n \in \mathbb{N}_0, \epsilon_i \in \{\pm 1\}$ and $x_1, \dots, x_n \in X$.

Rem 1.11 If M is a monoid and $x \in M$, then $\varphi : \mathbb{N}_0 \to M, n \mapsto x^n$ defines a monoid homomorphism. If G is a group and $x \in G$, one obtains a group homomorphism $\varphi : \mathbb{Z} \to G, n \mapsto x^n$, where for $n < 0$ we set $x^n = (x^{-n})^{-1}$.

Rem 1.27 $\mathrm{im}(f) \subset G'$ is a subgroup, but in general not a normal subgroup.

Def 2.3 An ideal $\mathfrak{a}$ in a ring R is a subset $\mathfrak{a} \subset R$ such that
(i) $(\mathfrak{a}, +) \subset (R, +)$ is a subgroup,
(ii) $r \in R, a \in \mathfrak{a} \Rightarrow ra \in \mathfrak{a}$.
The quotient group $R/\mathfrak{a}$ becomes a ring with $(a + \mathfrak{a})(b + \mathfrak{a}) = ab + \mathfrak{a}$.

Ex 2.2 $R = K$ a field; $R = \mathbb{Z}$; polynomial ring $R[T]$; the zero ring 0 (the unique ring up to isomorphism with $0 = 1$).

Rem 1.18 The left cosets of H in G are the equivalence classes with respect to $a \sim_H b \Leftrightarrow b^{-1}a \in H$. The elements of aH are called its representatives. The set of left cosets is denoted by G/H. Analogously, one defines the set $H \backslash G$ of right cosets $Ha := \{ha \mid h \in H\}$.

Lem 1.23 The following statements are equivalent:
(i) $H \subset G$ is a normal subgroup.
(ii) $aHa^{-1} = H \quad \forall a \in G$.
(iii) $aHa^{-1} \subset H \quad \forall a \in G$.

Rem 1.24 The subgroups $\{1\}$ and G are always normal subgroups in G for trivial reasons.

Def 1.28 Let $H \triangleleft G$. Then G/H with the operation $(aH)(a'H) = aa'H$ is called the quotient group of G by H.

Thm 1.16 Any two left cosets of H in G have the same cardinality, and distinct left cosets of H in G are disjoint. In particular, G is the disjoint union of the left cosets of H.

Lem 1.25 Every subgroup $H \subset G$ of index 2 is a normal subgroup.

Def 1.45 The order $\mathrm{ord}(a)$ of $a \in G$ is the smallest positive integer such that $a^{\mathrm{ord}(a)} = e$. If no such number exists, one sets $\mathrm{ord}(a) = \infty$.

Lem 1.33 If one of the subgroups H_1, H_2 is normal, then H_1H_2 is a subgroup. If both are normal, so is H_1H_2.

Thm 1.12 (Cayley's Theorem). Let G be a finite group with n elements. Then there exists an injective group homomorphism $\varphi : G \hookrightarrow \mathfrak{S}_n$.

Rem 2.13 The Axiom of Choice states: Every surjection of sets $f : M \to N$ splits, i.e., there exists $s : N \to M$ with $f \circ s = \mathrm{id}_N$.

Lem 1.41 Let $x \in G$. Then $\langle x \rangle$ is the image of the group homomorphism $\mathbb{Z} \to G, n \mapsto x^n$.

Def 2.50 Let R be a ring and $(M, +)$ a commutative monoid. Then $R[M] = R^{(M)} = \{(r_m)_{m \in M} \mid r_m = 0 \text{ for almost all } m \in M\}$ with componentwise addition and convolution multiplication $t_m = \sum_{m_1 + m_2 = m} r_{m_1} s_{m_2}$ is called the polynomial ring over M with coefficients in R (with basis X^m and $X^{m_1}X^{m_2} = X^{m_1 + m_2}$).

Def 2.6 $x \in R$ is called a zero divisor if there exists $y \in R, y \neq 0$ with $xy = 0$. R is called an integral domain (zero-divisor-free) if $R \neq 0$ and 0 is the only zero divisor. $x \in R$ is a unit if there exists $y \in R$ with $xy = 1$ (unit group $R^\times$). $\mathfrak{a} = (x) = Rx$ is called a principal ideal. R is a principal ideal domain (PID) if R is an integral domain and every ideal in R is principal.

Def 2.10 An ideal $\mathfrak{p} \subset R$ is called a prime ideal if $\mathfrak{p} \neq R$ and $xy \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. An ideal $\mathfrak{m}$ is called a maximal ideal if $\mathfrak{m} \neq R$ and there is no ideal $\mathfrak{a}$ with $\mathfrak{m} \subsetneq \mathfrak{a} \subsetneq R$.

Ex 2.4 $\mathfrak{a} = R$ and $\mathfrak{a} = \{0\}$ are ideals. If $\mathfrak{a}, \mathfrak{b}$ are ideals, so are $\mathfrak{a} \cap \mathfrak{b}$ and $\mathfrak{a} + \mathfrak{b}$. The subgroups of $\mathbb{Z}$ (0 and $n\mathbb{Z}$) are all ideals.

Lem 1.19 The bijective map $G \to G, a \mapsto a^{-1}$ defines a bijection $G/H \xrightarrow{\sim} H \backslash G$.

Thm 1.30 Let $H \triangleleft G$ and $\phi : G \to G/H$ be the canonical projection. Then $U \mapsto \phi^{-1}(U)$ induces an inclusion-preserving bijection between subgroups in G/H and subgroups in G containing H. U is normal in G/H if and only if $\phi^{-1}(U)$ is normal in G.

Lem 1.26 If $f : G \to G'$ is a group homomorphism, then $\ker(f) \subset G$ is a normal subgroup.

Thm 1.21 (Lagrange's Theorem). If $H \subset G$ is a subgroup, then $\mathrm{ord}(G) = \mathrm{ord}(H)(G : H)$.

Lem Zorn (Zorn's Lemma). Let $(M, \leq)$ be a non-empty partially ordered set. If every chain in M has an upper bound, then M contains at least one maximal element.

Lem 1.46 $\mathrm{ord}(a) = \mathrm{ord}(\langle a \rangle)$.

Thm 1.31 (Fundamental Theorem on Homomorphisms). Let $f : G \to G'$ be a group homomorphism. Then f factors uniquely in the form $G \xrightarrow{p} G/\ker(f) \xrightarrow{\sim} \mathrm{im}(f) \hookrightarrow G'$.

Lem 1.34 If $N \triangleleft G$ is a normal subgroup and $H \subset G$ is a subgroup, then N is normal in HN and $H \cap N$ is normal in H.

Thm 2.52 (Universal Property of the Polynomial Ring). Let $\phi : R \to R'$ be a ring homomorphism and $\sigma : M \to (R', \cdot)$ a monoid homomorphism. Then there exists a unique ring homomorphism $\Phi : R[M] \to R'$ with $\Phi|_R = \phi$ and $\Phi(X^m) = \sigma(m)$ for all $m \in M$.

Lem 2.31 If R is an integral domain, then for $f, g \in R[T]$: $\deg(f \cdot g) = \deg(f) + \deg(g)$.

Def 2.16 Let R be an integral domain. An element $0 \neq \pi \in R \setminus R^\times$ is called a prime element if (π) is a prime ideal ($\pi \mid ab \Rightarrow \pi \mid a$ or $\pi \mid b$); irreducible if $ab = \pi$ implies $a \in R^\times$ or $b \in R^\times$. Elements $a, b \in R$ are called associated ($a \doteq b$) if $a \mid b$ and $b \mid a$.

Lem 2.27 Let R be a principal ideal domain. Then every ascending chain $\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \mathfrak{a}_3 \subseteq \dots$ of ideals becomes stationary.

Def 2.23 An integral domain R is called a Euclidean domain if there is a map $v : R \setminus \{0\} \to \mathbb{N}_0$ such that for any $f, g \in R, g \neq 0$ there always exist $q, r \in R$ with $f = qg + r$ and ($v(r) < v(g)$ or $r = 0$).

Def 2.8 R is called a field if $R^\times = R \setminus \{0\}$ (in particular, $R = 0$ is not a field).

Thm 2.5 The assignment $\mathfrak{b} \mapsto \phi^{-1}(\mathfrak{b})$ under $\phi : R \to R/\mathfrak{a}$ defines an inclusion-preserving bijection between ideals in $R/\mathfrak{a}$ and ideals in R containing $\mathfrak{a}$.

Rem 1.29 The canonical projection $G \to G/H, g \mapsto gH$ is a group homomorphism with kernel H. Thus normal subgroups are precisely the subgroups that occur as kernels of group homomorphisms.

Thm 2.14 Every ring $R \neq 0$ possesses a maximal ideal.

Cor 1.47 For $a \in G$, $\mathrm{ord}(a) \mid \mathrm{ord}(G)$.

Thm 1.36 (Second Isomorphism Theorem). Let G be a group and $N, H \triangleleft G$ with $N \subset H$. Then there is a natural isomorphism $(G/N)/(H/N) \simeq G/H$.

Thm 1.37 The subgroups of $\mathbb{Z}$ are precisely $\{0\}$ and $n\mathbb{Z}$ ($n \in \mathbb{N}$). For $n \in \mathbb{N}$, $n\mathbb{Z}$ is isomorphic to $\mathbb{Z}$.

Thm 1.35 (First Isomorphism Theorem). Let G be a group, $H \subset G$ a subgroup, and $N \triangleleft G$ a normal subgroup. Then the natural inclusion $H \hookrightarrow HN$ induces an isomorphism $H/(H \cap N) \simeq HN/N$.

Def 1.42 A group G is called cyclic if it is generated by one element. Equivalently: there exists a surjective group homomorphism $\mathbb{Z} \to G$.

Cor 2.53 Given a ring homomorphism $\phi : R \to R'$ and $x_1, \dots, x_n \in R'$, there exists a unique ring homomorphism $\Phi : R[X_1, \dots, X_n] \to R'$ with $\Phi|_R = \phi$ and $\Phi(X_i) = x_i$ ($i = 1, \dots, n$).

Lem 2.17 For $a, b \in R$, the following are equivalent:
(i) $a \doteq b$.
(ii) $a = be$ with $e \in R^\times$.
(iii) $(a) = (b)$.

Rem 2.28 A ring in which every ascending chain of ideals becomes stationary is called Noetherian.

Ex 2.24 $R = k[T]$ (k field) with $v(f) = \deg(f)$; $R = \mathbb{Z}$ with $v(a) = |a|$.

Def 3.4 Let K be a field. The intersection of all subfields of K is called the prime field of K.

Thm 2.9 Let R be a ring $\neq 0$. Then the following are equivalent:
(i) R is a field.
(ii) (0) and (1) are the only ideals in R.
(iii) Every homomorphism $f : R \to S$ into a ring $S \neq 0$ is injective.

Ex 2.7 $\mathbb{Z}$ is a principal ideal domain.

Def 3.1 Let R be an integral domain and $\varphi : \mathbb{Z} \to R, n \mapsto n \cdot 1_R$. The uniquely determined $n \in \mathbb{N}_0$ with $(n) = \ker(\varphi)$ is called the characteristic $\mathrm{char}(R)$ of R ($\mathrm{char}(R) = 0$ or p prime).

Cor 2.15
(i) Every proper ideal $\mathfrak{a} \subsetneq R$ is contained in a maximal ideal.
(ii) Every non-unit is contained in a maximal ideal.

Thm 1.44 If G is a cyclic group, then every subgroup and every quotient group of G is again cyclic.

Cor 1.43 Let G be a cyclic group. Then $G \cong \mathbb{Z}$ if $\mathrm{ord}(G) = \infty$, and $G \cong \mathbb{Z}/n\mathbb{Z}$ if $\mathrm{ord}(G) = n \in \mathbb{N}$.

Def 2.54 Let $R \subset R'$ be a ring extension. A system $(x_1, \dots, x_n)$ of elements in R' is called algebraically independent (or transcendental) over R if the evaluation homomorphism $R[X_1, \dots, X_n] \to R', X_i \mapsto x_i$ is injective; otherwise algebraically dependent.

Lem 2.19 Suppose $a \in R$ has factorizations $a = p_1 \cdots p_r = q_1 \cdots q_s$ into prime elements p_i and irreducible elements q_j. Then $r = s$ and after renumbering, $p_i \doteq q_i$ ($i = 1, \dots, r$).

Def 2.33 Let R be an integral domain and $M = \{(a, b) \mid a, b \in R, b \neq 0\}$ with $(a_1, b_1) \sim (a_2, b_2) \Leftrightarrow a_1b_2 = a_2b_1$. The set of equivalence classes $Q(R)$ with $\frac{a_1}{b_1} + \frac{a_2}{b_2} = \frac{a_1b_2 + a_2b_1}{b_1b_2}$ and $\frac{a_1}{b_1} \cdot \frac{a_2}{b_2} = \frac{a_1a_2}{b_1b_2}$ is a field, called the field of fractions of R.

Thm 2.25 Every Euclidean domain is a principal ideal domain.

Rem 3.5 The prime field is the smallest subfield of K.

Lem 3.3 If K, L are fields with $\mathrm{char}(K) \neq \mathrm{char}(L)$, there is no field homomorphism from K to L.

Def 3.7 Let K be a field with $\mathrm{char}(K) = p > 0$. Then the field homomorphism $\sigma : K \to K, a \mapsto a^p$, is called the Frobenius homomorphism of K.

Thm 2.11
(i) $\mathfrak{p}$ is a prime ideal $\Leftrightarrow R/\mathfrak{p}$ is an integral domain.
(ii) $\mathfrak{m}$ is a maximal ideal $\Leftrightarrow R/\mathfrak{m}$ is a field.
In particular, maximal ideals are prime.

Ex 3.2 $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic 0; $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ and $\mathbb{F}_p[T]$ have characteristic p; $\mathrm{char}(R) = \mathrm{char}(Q(R))$; for subrings $R \subset S$, $\mathrm{char}(R) = \mathrm{char}(S)$.

Thm 1.49 Let G be a finite group of prime order, i.e., $\mathrm{ord}(G) = p$, p prime. Then G is cyclic and $G \cong \mathbb{Z}/p\mathbb{Z}$.

Def 3.15 $\alpha \in L$ is called algebraic over K if $\alpha^n + c_{n-1}\alpha^{n-1} + \dots + c_0 = 0$ holds with $c_i \in K$. L is called algebraic over K if every element of L is algebraic over K.

Ex 2.55 Over $\mathbb{Q}$ in $\mathbb{C}$: $(\sqrt{2})$ is algebraically dependent ($X^2 - 2 = 0$); e is transcendental (Hermite); (e, e^2) is algebraically dependent ($X^2 - Y = 0$); π is transcendental (Lindemann); whether (e, π) is transcendental is an open problem.

Thm 2.20 Let R be an integral domain. Then the following are equivalent:
(i) Every $a \in R \setminus R^\times, a \neq 0$, can be written uniquely (up to order and association) as a product of irreducible elements.
(ii) Every $a \in R \setminus R^\times, a \neq 0$, can be written as a product of prime elements.

Rem 2.34 The natural map $R \to Q(R), x \mapsto \frac{x}{1}$ is an injective ring homomorphism.

Lem 2.18 Prime elements are irreducible. In a principal ideal domain, every irreducible element is prime.

Rem 3.8 σ is an injective field homomorphism because $(a+b)^p = a^p + b^p$ in characteristic p.

Lem 3.6 Let K be a field and $P \subset K$ its prime field. Then $\mathrm{char}(K) = 0 \Leftrightarrow P \cong \mathbb{Q}$ and $\mathrm{char}(K) = p > 0 \Leftrightarrow P \cong \mathbb{F}_p$.

Ex 2.12 For $f : \mathbb{Z} \to \mathbb{Q}$, $(0) \subset \mathbb{Q}$ is a maximal ideal. $f^{-1}((0)) = (0)$ is a prime ideal in $\mathbb{Z}$, but not maximal.

Def/Lem 3.16 Let $K \subset L$ be a field extension and $\alpha \in L$ algebraic over K. Then there exists a unique monic polynomial of minimal degree $f \in K[X]$ with $f(\alpha) = 0$. We have $\ker(\varphi) = (f)$ for $\varphi : K[X] \to L, g \mapsto g(\alpha)$, and f is irreducible. f is called the minimal polynomial of α.

Ex 3.15a $\mathbb{C}/\mathbb{R}$ is algebraic; $k(t)/k$ is not algebraic.

Def 2.21 An integral domain satisfying the equivalent conditions of Theorem 2.20 is called a unique factorization domain (UFD). In a UFD, gcd and lcm exist up to association.

Thm 2.37 (Universal Property of $Q(R)$). Let R be an integral domain and $f : R \hookrightarrow K$ an injective ring homomorphism into a field K. Then there exists a unique field homomorphism $\phi : Q(R) \hookrightarrow K$ extending f.

Thm 2.26 Every principal ideal domain is a unique factorization domain.

Cor 2.32 If R is an integral domain, so is $R[T]$.

Lem 3.9 If K is a finite field, the Frobenius homomorphism is an automorphism.

Ex 2.51 $R[\mathbb{N}_0] \cong R[T]$; $R[(\mathbb{N}_0)^n] \cong R[X_1, \dots, X_n]$; $R[(\mathbb{N}_0)^{(I)}] \cong R[(X_i)_{i \in I}]$; for an abelian group G, $R[G]$ is called the group ring.

Rem 2.38 If R is a UFD and $(p_i)_{i \in I}$ is a system of representatives of the primes, every $\frac{a}{b} \in Q(R)^\times$ has a unique representation $\frac{a}{b} = \epsilon \prod_{i \in I} p_i^{v_i(a/b)}$ with $v_i(\frac{a}{b}) \in \mathbb{Z}$. For $f = \sum a_iT^i \in Q(R)[T]$ and prime $p \in R$, set $v_p(f) = \min_i v_p(a_i)$. Then $f \in R[T] \Leftrightarrow v_p(f) \geq 0 \ \forall p$.

Cor 2.22 In a unique factorization domain, every irreducible element is a prime element.

Cor 2.35 Let R be an integral domain and $f \in R[T], f \neq 0$. Then f has at most $\deg(f)$ roots in R.

Lem 2.29 If R is a principal ideal domain, the ideal identities $(a) + (b) = (\gcd(a,b))$ and $(a) \cap (b) = (\mathrm{lcm}(a,b))$ hold.

Thm 1.48 (Fermat's Little Theorem). If G is a finite group, then $a^{\mathrm{ord}(G)} = e$ holds for all $a \in G$.

Def 2.42 Let R be a UFD. A polynomial $f \in R[T]$ is called primitive if the gcd of its coefficients is 1, i.e., $v_p(f) = 0$ for all prime elements $p \in R$.

Ex 2.39 In $\mathbb{Z}$: $v_3(\frac{2}{9}) = -2, v_2(\frac{2}{9}) = 1$, and $v_p(\frac{2}{9}) = 0$ for every prime $p \neq 2, 3$.

Thm 2.40 (Gauss's Lemma). Let R be a UFD and $p \in R$ a prime element. Then for $f, g \in Q(R)[T]$: $v_p(fg) = v_p(f) + v_p(g)$.

Ex 2.36 In the (non-domain) ring $R = \mathbb{Z}/8\mathbb{Z}$, the quadratic polynomial $T^2 - 1$ has four roots: $1, 3, 5, 7$.

Def 3.17 Let $K \subset L$ be a field extension and $\alpha \in L$. $K[\alpha] = \{\sum_{i=0}^n c_i\alpha^i \mid c_i \in K\}$ is the smallest subring of L containing K and α. $K(\alpha) = Q(K[\alpha]) \subset L$ is the smallest subfield of L containing K and α (K adjoined α). More generally, $K[S]$ and $K(S)$ are defined for $S \subset L$.

Def 3.11 Let $K \subset L$ be fields (L extension field of K). $\dim_K L$ is called the degree of the field extension L over K, denoted $[L : K] \in \mathbb{N} \cup \{\infty\}$.

Thm 2.30 If R is a principal ideal domain, every non-zero prime ideal is maximal.

Lem 3.10 Let $\mathrm{char}(K) = p > 0$ and $P \subset K$ be the prime field. Then $P = \{a \in K \mid \sigma(a) = a\}$.

Ex 2.43 Every monic polynomial is primitive.

Lem 2.44 Let $0 \neq f \in Q(R)[T]$. Then there exist $0 \neq a \in Q(R)$ and a primitive $\tilde{f} \in R[T]$ such that $f = a\tilde{f}$.

Cor 2.41 Let R be a UFD and $h \in R[T]$ a monic polynomial. If $h = fg$ with monic polynomials $f, g \in Q(R)[T]$, then $f, g \in R[T]$.

Def 3.22 Let $K \subset L$ and $\alpha \in L$. $[K(\alpha) : K]$ is called the degree of α over K. L/K is called simple if $L = K(\alpha)$ for some $\alpha \in L$. L/K is called finitely generated if $L = K(\alpha_1, \dots, \alpha_n)$ for finitely many $\alpha_i \in L$.

Ex 3.14a $[\mathbb{C} : \mathbb{R}] = 2$; $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$; $[\mathbb{R} : \mathbb{Q}] = \infty$; $[k(t) : k] = \infty$.

Rem 3.12 $[L : K] = 1 \Leftrightarrow L = K$.

Thm 3.13 (Tower Law / Degree Theorem). Let $K \subset L \subset M$ be field extensions. Then $[M : K] = [M : L] \cdot [L : K]$.

Thm 3.18 Let $K \subset L$ be a field extension and $\alpha \in L$ algebraic with minimal polynomial f over K. Then $K[\alpha] = K(\alpha)$ is a field, φ induces $K[X]/(f) \xrightarrow{\sim} K(\alpha)$, and $[K(\alpha) : K] = \deg(f) < \infty$.

Thm 3.19 Every finite field extension is algebraic.

Def 3.31 A field K is called algebraically closed if every $f \in K[X]$ with $\deg(f) \geq 1$ has a root in K.

Thm 3.30 Let K be a field and $f \in K[X]$ with $\deg(f) \geq 1$. Then there exists a finite (algebraic) field extension L/K such that f has a root in L.

Thm 2.45 (Gauss's Theorem). Let R be a UFD. Then $R[T]$ is also a UFD. A polynomial $q \in R[T]$ is a prime element in $R[T]$ if and only if:
(i) $q \in R$ and q is a prime element in R, or
(ii) q is primitive in $R[T]$ and a prime element in $Q(R)[T]$.

Ex 3.22a $\mathbb{C}/\mathbb{R} = \mathbb{R}(i)/\mathbb{R}$ and $k(t)/k$ are simple. $k(X, Y)/k$ is finitely generated, but not simple.

Cor 3.14 If $K \subset L \subset M$ are field extensions and $[M : K]$ is a prime number, then $L = K$ or $L = M$.

Cor 3.21 Let $K \subset L$ and $\alpha \in L$ be algebraic over K. Then the field extension $K(\alpha)/K$ is algebraic.

Rem 3.20 There exist infinite algebraic extensions.

Lem 3.33 K is algebraically closed if and only if there is no proper ($L \neq K$) finite extension L/K.

Rem 3.32 Over an algebraically closed field K, every $0 \neq f \in K[X]$ splits into linear factors $f = c(X - \alpha_1) \cdots (X - \alpha_n)$, and every irreducible polynomial has degree 1.

Thm 3.34 For every field K, there exists an algebraically closed extension field L.

Thm 2.47 (Eisenstein's Irreducibility Criterion). Let R be a UFD and $f = a_nT^n + \dots + a_0 \in R[T]$ be primitive of degree > 0. Let $p \in R$ be a prime element with $p \nmid a_n$, $p \mid a_i$ for $i < n$, and $p^2 \nmid a_0$. Then f is irreducible in $R[T]$ and in $Q(R)[T]$.

Thm 2.48 (Reduction Criterion). Let R be a UFD, $p \in R$ prime, and $f \in R[T]$ a polynomial whose leading coefficient is not divisible by p. If $\phi(f)$ under $\phi : R[T] \to (R/(p))[T]$ is irreducible, then f is irreducible in $Q(R)[T]$ (and in $R[T]$ if f is primitive).

Cor 2.46 Let R be a UFD. Then the polynomial ring in n variables $R[T_1, \dots, T_n]$ is a UFD (in particular $k[T_1, \dots, T_n]$ for a field k and $\mathbb{Z}[T_1, \dots, T_n]$).

Thm 3.23 Let $L = K(\alpha_1, \dots, \alpha_n)$ be finitely generated. If $\alpha_1, \dots, \alpha_n$ are algebraic over K, then:
(i) $L = K(\alpha_1, \dots, \alpha_n) = K[\alpha_1, \dots, \alpha_n]$.
(ii) L/K is finite, in particular algebraic.

Ex 2.47a $T^n - t \in k(t)[T]$ is irreducible ($p = t$). $T^3 + 5T^2 + 5 \in \mathbb{Q}[T]$ is irreducible ($p = 5$). For p prime, the cyclotomic polynomial $f(T) = T^{p-1} + \dots + 1 = \frac{T^p - 1}{T - 1}$ is irreducible in $\mathbb{Q}[T]$ (via Eisenstein applied to $f(T+1)$).

Ex 2.49a 1) $X^3 + 3X^2 - 4X - 1 \in \mathbb{Q}[X]$ is irreducible ($p = 3$).
2) $X^4 + 3X^3 + 5XY^2 + Y + 3 \in \mathbb{Q}[X, Y]$ is irreducible ($R = \mathbb{Q}[Y], p = Y$ and Eisenstein with $p = 3$).

Rem 2.49 Theorem 2.48 is mainly applied when R is a PID ($R/(p)$ is a field by Theorem 2.30) or for $R = k[T_1, \dots, T_n]$ with $p = T_n$.

Cor 3.26 Let $K \subset L$ be a field extension. Then the set $\{\alpha \in L \mid \alpha \text{ algebraic over } K\}$ is a subfield of L.

Thm 3.28 Let $K \subset L \subset M$ be field extensions. If $\alpha \in M$ is algebraic over L and L/K is algebraic, then α is algebraic over K. In particular, M/K is algebraic if and only if L/K and M/L are algebraic.

Cor 3.24 Let $K \subset L$ be a field extension. Then the following are equivalent:
(i) L/K is finite.
(ii) L is generated over K by finitely many algebraic elements.
(iii) L is a finitely generated algebraic field extension of K.

Ex 3.29 For $\mathbb{Q}^{\mathrm{alg}} \subset \mathbb{C}$, $\mathbb{Q}^{\mathrm{alg}}/\mathbb{Q}$ is algebraic with $[\mathbb{Q}^{\mathrm{alg}} : \mathbb{Q}] = \infty$ (by Eisenstein). Since $\mathbb{Q}^{\mathrm{alg}}$ is countable and $\mathbb{C}$ is uncountable, there are uncountably many transcendental elements in $\mathbb{C}$.

Def 3.27 The subfield $\{\alpha \in L \mid \alpha \text{ algebraic over } K\}$ is called the (relative) algebraic closure of K in L.

Cor 3.25 Let $K \subset L$ be a field extension. Then the following are equivalent:
(i) L/K is algebraic.
(ii) L/K is generated by algebraic elements.